

9. Compare Linear and Polynomial Regression using Python

AIM

To compare the performance of **Linear Regression** and **Polynomial Regression** using Python.

ALGORITHM

1. Import the required libraries.
2. Create a sample dataset.
3. Train the Linear Regression model.
4. Transform the data into polynomial features.
5. Train the Polynomial Regression model.
6. Predict the output using both models.
7. Compare the performance using R^2 Score.
8. Plot the Linear and Polynomial Regression curves.

PROGRAM:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import r2_score

X = np.array([[1],[2],[3],[4],[5],[6],[7],[8],[9],[10]])
y = np.array([2,5,10,17,26,37,50,65,82,101])

linear_model = LinearRegression()
linear_model.fit(X, y)
y_linear = linear_model.predict(X)
```

```
poly = PolynomialFeatures(degree=2)
```

```
X_poly = poly.fit_transform(X)
```

```
poly_model = LinearRegression()
```

```
poly_model.fit(X_poly, y)
```

```
y_poly = poly_model.predict(X_poly)
```

```
print("Linear Regression R2 Score:", r2_score(y, y_linear))
```

```
print("Polynomial Regression R2 Score:", r2_score(y, y_poly))
```

```
plt.scatter(X, y, color='blue')
```

```
plt.plot(X, y_linear, color='red', label='Linear Regression')
```

```
plt.plot(X, y_poly, color='green', label='Polynomial Regression')
```

```
plt.xlabel("X")
```

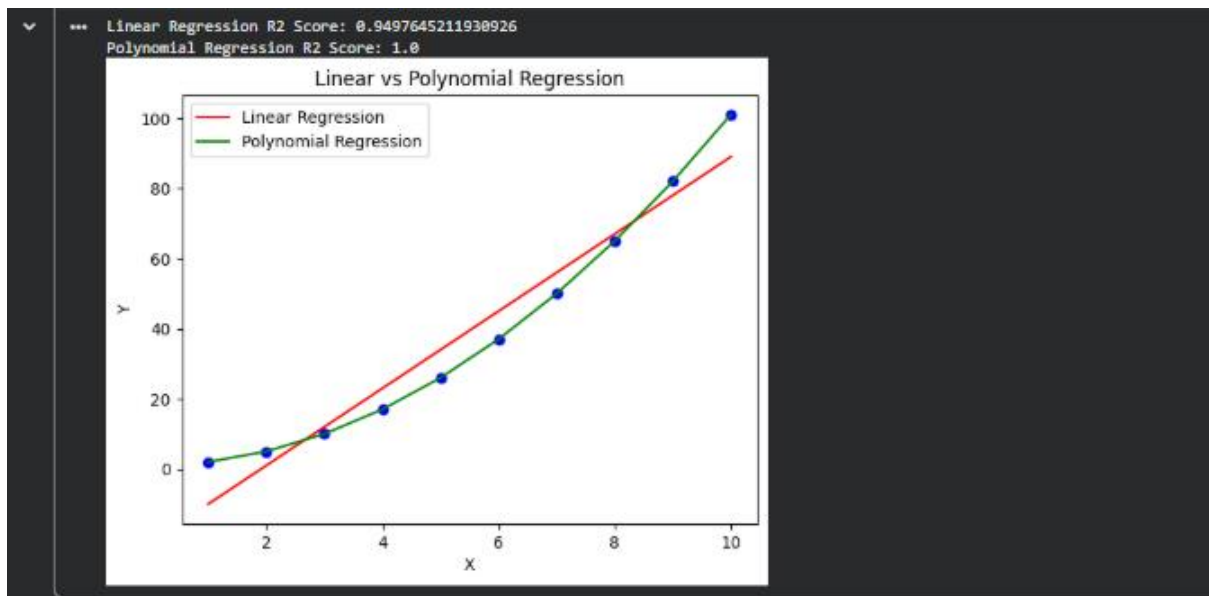
```
plt.ylabel("Y")
```

```
plt.title("Linear vs Polynomial Regression")
```

```
plt.legend()
```

```
plt.show()
```

OUTPUT:



A graph will be displayed showing:

- **Red Line** → Linear Regression
- **Green Curve** → Polynomial Regression

RESULT:

The **Linear Regression** and **Polynomial Regression** models were successfully implemented and compared. The Polynomial Regression model achieved a higher **R² Score** and provided a better fit for the nonlinear dataset than the Linear Regression model.